

Questions

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Q1 - 2024 (04 Apr Shift 2)

Let $f(x) = 3\sqrt{x-2} + \sqrt{4-x}$ be a real valued function. If α and β are respectively the minimum and the maximum values of f , then $\alpha^2 + 2\beta^2$ is equal to

- (1) 42
- (2) 38
- (3) 24
- (4) 44

Q2 - 2024 (04 Apr Shift 2)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function such that $f(0) = 0$, $f(1) = 1$, $f(2) = -1$, $f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeros of $(3f'f'' + ff''')(x)$ is _____

Q3 - 2024 (05 Apr Shift 1)

Let a rectangle $ABCD$ of sides 2 and 4 be inscribed in another rectangle $PQRS$ such that the vertices of the rectangle $ABCD$ lie on the sides of the rectangle $PQRS$. Let a and b be the sides of the rectangle $PQRS$ when its area is maximum. Then $(a+b)^2$ is equal to :

- (1) 72
- (2) 60
- (3) 64
- (4) 80

Q4 - 2024 (05 Apr Shift 1)

For the function

$$f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x), \text{ where } x \in \left[0, \frac{\pi}{2}\right],$$

consider the following two statements :

- (I) f is increasing in $\left(0, \frac{\pi}{2}\right)$.

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(II) f' is decreasing in $(0, \frac{\pi}{2})$.

Between the above two statements,

(1) only (II) is true.

(2) only (I) is true.

(3) neither (I) nor (II) is true.

(4) both (I) and (II) are true

Q5 - 2024 (05 Apr Shift 2)

Let the maximum and minimum values of $(\sqrt{8x - x^2} - 12 - 4)^2 + (x - 7)^2, x \in \mathbf{R}$ be M and m , respectively. Then $M^2 - m^2$ is equal to _____

Q6 - 2024 (06 Apr Shift 1)

The interval in which the function $f(x) = x^x, x > 0$, is strictly increasing is

(1) $(0, \frac{1}{e}]$

(2) $(0, \infty)$

(3) $[\frac{1}{e}, \infty)$

(4) $[\frac{1}{e^2}, 1)$

Q7 - 2024 (06 Apr Shift 2)

If the function $f(x) = (\frac{1}{x})^{2x}; x > 0$ attains the maximum value at $x = \frac{1}{e}$ then :

(1) $e^\pi < \pi^e$

(2) $e^\pi > \pi^e$

(3) $(2e)^\pi > \pi^{(2e)}$

(4) $e^{2\pi} < (2\pi)^e$

Q8 - 2024 (08 Apr Shift 1)

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For the function $f(x) = (\cos x) - x + 1$, $x \in \mathbb{R}$, between the following two statements

(S1) $f(x) = 0$ for only one value of x in $[0, \pi]$.

(S2) $f(x)$ is decreasing in $\left[0, \frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2}, \pi\right]$.

(1) Both (S1) and (S2) are correct.

(2) Both (S1) and (S2) are incorrect.

(3) Only (S2) is correct.

(4) Only (S1) is correct.

Q9 - 2024 (08 Apr Shift 1)

Let $f(x) = 4 \cos^3 x + 3\sqrt{3} \cos^2 x - 10$. The number of points of local maxima of f in interval $(0, 2\pi)$ is

(1) 3

(2) 4

(3) 1

(4) 2

Q10 - 2024 (08 Apr Shift 1)

The number of critical points of the function $f(x) = (x - 2)^{2/3}(2x + 1)$ is

(1) 1

(2) 2

(3) 0

(4) 3

Q11 - 2024 (08 Apr Shift 2)

If the function $f(x) = 2x^3 - 9x^2 + 12ax + 1$, $a > 0$ has a local maximum at $x = \alpha$ and a local minimum at $x = \alpha^2$, then α and α^2 are the roots of the equation :

(1) $x^2 - 6x + 8 = 0$

(2) $x^2 + 6x + 8 = 0$

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$$(3) 8x^2 + 6x - 1 = 0$$

$$(4) 8x^2 - 6x + 1 = 0$$

Q12 - 2024 (08 Apr Shift 2)

Let A be the region enclosed by the parabola $y^2 = 2x$ and the line $x = 24$. Then the maximum area of the rectangle inscribed in the region A is _____

Q13 - 2024 (09 Apr Shift 1)

A variable line L passes through the point $(3, 5)$ and intersects the positive coordinate axes at the points A and B. The minimum area of the triangle OAB, where O is the origin, is :

$$(1) 30$$

$$(2) 25$$

$$(3) 40$$

$$(4) 35$$

Q14 - 2024 (09 Apr Shift 1)

Let the set of all positive values of λ , for which the point of local minimum of the function $(1 + x(\lambda^2 - x^2))$ satisfies $\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$, be (α, β) . Then $\alpha^2 + \beta^2$ is equal to _____

Q15 - 2024 (09 Apr Shift 2)

Let the set of all values of p , for which $f(x) = (p^2 - 6p + 8)(\sin^2 2x - \cos^2 2x) + 2(2 - p)x + 7$ does not have any critical point, be the interval (a, b) . Then $16ab$ is equal to _____

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Answer Key

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Solutions

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Q1

$$f(x) = 3\sqrt{x-2} + \sqrt{4-x}$$

$$x-2 \geq 0 \& 4-x \geq 0$$

$$\therefore x \in [2, 4]$$

$$\text{Let } x = 2 \sin^2 \theta + 4 \cos^2 \theta$$

$$\therefore f(x) = 3\sqrt{2}|\cos \theta| + \sqrt{2}|\sin \theta|$$

$$\therefore \sqrt{2} \leq 3\sqrt{2}|\cos \theta| + \sqrt{2}|\sin \theta| \leq \sqrt{9 \times 2 + 2}$$

$$\sqrt{2} \leq 3\sqrt{2}|\cos \theta| + \sqrt{2}|\sin \theta| \leq \sqrt{20}$$

$$\therefore \alpha = \sqrt{2} \quad \beta = \sqrt{20}$$

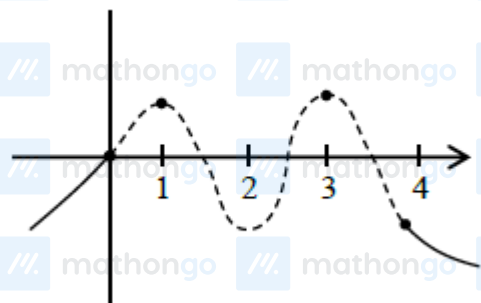
$$\alpha^2 + 2\beta^2 = 2 + 40 = 42$$

Q2

$$(3f'f'' + ff''')(x) = \left((ff'' + (f')^2)(x) \right)'$$

$$(ff'f'' + (f')^2)(x) = ((ff')(x))'$$

$$\therefore (3f'f'' + f''')(x) = (f(x) \cdot f'(x))''$$



$$\text{min roots of } f(x) \rightarrow 4$$

$$\therefore \text{min. roots of } f'(x) \rightarrow 3$$

$$\therefore \text{min roots of } (f(x) \cdot f'(x)) \rightarrow 7$$

$$\therefore \text{min roots of } (f(x) \cdot f'(x))'' \rightarrow 5$$

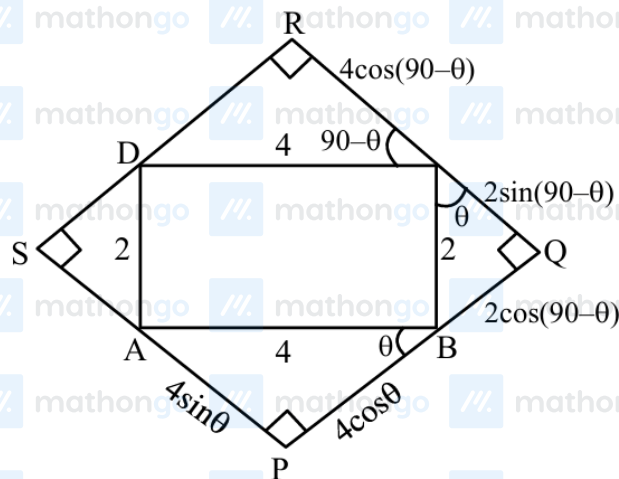
Q3

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Solutions

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$$\text{Area} = (4 \cos \theta + 2 \sin \theta)(2 \cos \theta + 4 \sin \theta)$$

$$= 8 \cos^2 \theta + 16 \sin \theta \cos \theta + 4 \sin \theta \cos \theta + 8 \sin^2 \theta$$

$$= 8 + 20 \sin \theta \cos \theta$$

$$= 8 + 10 \sin 2\theta$$

$$\text{Max Area} = 8 + 10 = 18 (\sin 2\theta = 1) \theta = 45^\circ$$

$$(a+b)^2 = (4 \cos \theta + 2 \sin \theta + 2 \cos \theta + 4 \sin \theta)^2$$

$$= (6 \cos \theta + 6 \sin \theta)^2$$

$$= 36(\sin \theta + \cos \theta)^2$$

$$= 36(\sqrt{2})^2$$

$$= 72$$

Q4

$$f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x) \quad x \in \left[0, \frac{\pi}{2}\right]$$

$$f'(x) = \cos x + 3 - \frac{2}{\pi}(2x + 1) > 0 \quad f(x) \uparrow$$

$$f'(x) = -\sin x + 0 - \frac{\pi}{2}(2)$$

$$= -\sin x - \frac{4}{\pi} < 0 \quad f'(x) \downarrow$$

$$0 < x < \frac{\pi}{2}$$

$$\Rightarrow -\frac{2}{\pi} \left(\begin{matrix} 0 < 2x < \pi \\ +1 & +1 & +1 \end{matrix} \right)$$

$$-\frac{2}{\pi} > -\frac{2}{\pi} \left(\begin{matrix} 2x+1 \\ +3 \end{matrix} \right) > -\frac{2}{\pi} \left(\begin{matrix} \pi+1 \\ +3 \end{matrix} \right)$$

$$\begin{matrix} +3 \\ 3 \end{matrix} - \frac{2}{\pi} > 3 - \frac{2}{\pi}(2x+1) > 3 - \frac{2}{\pi} \left(\begin{matrix} \pi+1 \\ +ve \end{matrix} \right)$$

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Solutions

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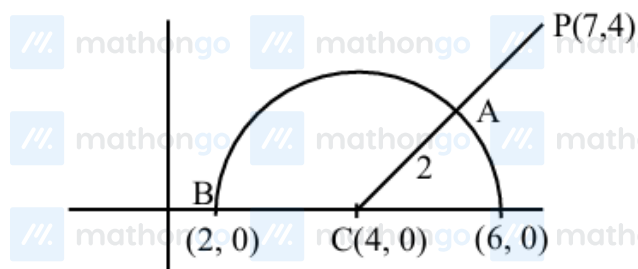
Q5

$$(x-7)^2 + (y-4)^2$$

$$y = \sqrt{8x - x^2 - 12}$$

$$y^2 = -(x-4)^2 + 16 - 12$$

$$(x-4)^2 + y^2 = 4$$



$$m = 9$$

$$M = 41$$

$$M^2 - m^2 = 41^2 - 9^2 = 1600$$

Q6

$$f(x) = x^x; x > 0$$

$$\ell n n = x \ell n x$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{x}{x} + \ln x$$

$$\frac{dy}{dx} = x^x (1 + \ln x)$$

for strictly increasing

$$\frac{dy}{dx} \geq 0 \Rightarrow x^x (1 + \ln x) \geq 0$$

$$\Rightarrow \ln x \geq -1$$

$$x \geq e^{-1}$$

$$x \geq \frac{1}{e}$$

$$x \in \left[\frac{1}{e}, \infty \right)$$

Q7

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Solutions

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Let $y = \left(\frac{1}{x}\right)^{2x}$

$$\ell ny = 2x \ln \left(\frac{1}{x}\right)$$

$$\ell ny = -2x \ell nx$$

$$\frac{1}{y} \frac{dy}{dx} = -2(1 + \ell nx)$$

for $x > \frac{1}{e}$ f^n is decreasing

so, $e < \pi$

$$\left(\frac{1}{e}\right)^{2e} > \left(\frac{1}{\pi}\right)^{2\pi}$$

$$e^\pi > \pi^e$$

Q8

$$f'(x) = \cos x - x + 1$$

$$f'(x) = -\sin x - 1$$

f is decreasing $\forall x \in \mathbb{R}$

$$f(x) = 0$$

$$f(0) = 2, f(\pi) = -\pi$$

f is strictly decreasing in $[0, \pi]$ and $f(0) \cdot f(\pi) < 0$

$\Rightarrow$ only one solution of $f(x) = 0$

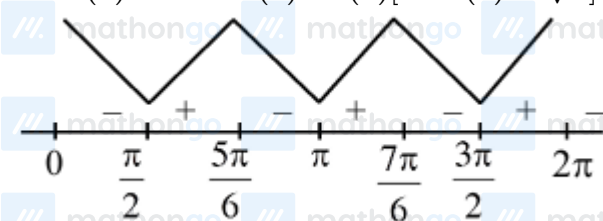
S1 is correct and S2 is incorrect.

Q9

$$f(x) = 4 \cos^3(x) + 3\sqrt{3} \cos^2(x) - 10; x \in (0, 2\pi)$$

$$\Rightarrow f'(x) = 12 \cos^2 x [-\sin(x)] + 3\sqrt{3}(2 \cos(x))[-\sin(x)]$$

$$\Rightarrow f'(x) = -6 \sin(x) \cos(x) [2 \cos(x) + \sqrt{3}]$$



local maxima at $x = \frac{5\pi}{6}, \frac{7\pi}{6}$

Q10

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Solutions

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$$f(x) = (x-2)^{2/3}(2x+1)$$

$$f'(x) = \frac{2}{3}(x-2)^{-1/3}(2x+1) + (x-2)^{2/3}$$

$$f'(x) = 2 \times \frac{(2x+1) + (x-2)}{3(x-2)^{1/3}}$$

$$\frac{3x-1}{(x-2)^{1/3}} = 0$$

Critical points $x = \frac{1}{3}$ and $x = 2$

Q11

$$f'(x) = 6x^2 - 18ax + 12a^2 = 0$$

$$\alpha + \alpha^2 = 3a \text{ \& } \alpha \times \alpha^2 = 2a^2$$

$$\downarrow$$

$$(\alpha + \alpha^2)^3 = 27a^3$$

$$\Rightarrow 2a^2 + 4a^4 + 3(3a)(2a^2) = 27a^3$$

$$\Rightarrow 2 + 4a^2 + 18a = 27a$$

$$\Rightarrow 4a^2 - 9a + 2 = 0$$

$$\Rightarrow 4a^2 - 8a - a + 2 = 0$$

$$\Rightarrow (4a-1)(a-2) = 0 \Rightarrow a = 2$$

$$\text{so } 6x^2 - 36x + 48 = 0$$

$$\Rightarrow x^2 - 6x + 8 = 0 \dots (1)$$

If we take $a = \frac{1}{4}$ then $\alpha = \frac{1}{2}$ which is not possible

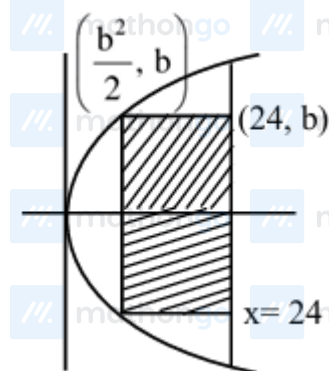
Q12

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Solutions

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$$A = 2 \left(24 - \frac{b^2}{2} \right) \cdot b$$

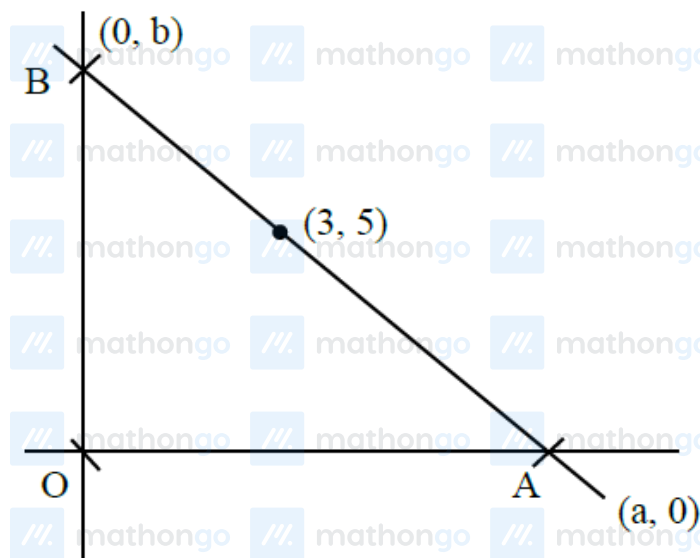
$$\frac{dA}{db} = 0 \Rightarrow b = 4$$

$$A = 2(24 - 8)4 = 128$$

Q13

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{3}{a} + \frac{5}{b} = 1 \Rightarrow b = \frac{5a}{a-3}, a > 3$$



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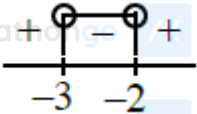
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Solutions

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$$\begin{aligned}
 A &= \frac{1}{2}ab = \frac{1}{2}a \frac{5a}{(a-3)} = \frac{5}{2} \cdot \frac{a^2}{a-3} \\
 &= \frac{5}{2} \left(\frac{a^2 - 9 + 9}{a-3} \right) \\
 &= \frac{5}{2} \left(a + 3 + \frac{9}{a-3} \right) \\
 &= \frac{5}{2} \left(a - 3 + \frac{9}{a-3} + 6 \right) \geq 30
 \end{aligned}$$

Q14

$$\begin{aligned}
 \frac{x^2 + x + 2}{x^2 + 5x + 6} &< 0 \\
 \Rightarrow \frac{1}{(x+2)(x+3)} &< 0
 \end{aligned}$$


$$x \in (-3, -2) \dots \dots \dots (1)$$

$$f(x) = 1 + x(\lambda^2 - x^2)$$

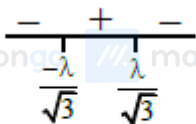
Finding local minima

$$f'(x) = (\lambda^2 - x^2) + (-2x) \cdot x$$

$$\text{Put } f'(x) = 0$$

$$\Rightarrow \lambda^2 = 3x^2$$

$$\Rightarrow x = \pm \frac{\lambda}{\sqrt{3}}$$



Local min

Local max

We want local min

$$\Rightarrow x = \frac{-\lambda}{\sqrt{3}}$$

from (1)

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Solutions

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$$x \in (-3, -2)$$

$$-3 < \frac{-\lambda}{\sqrt{3}} < -2$$

$$3\sqrt{3} > \lambda > 2\sqrt{3}$$

$$\alpha = 2\sqrt{3}, \beta = 3\sqrt{3}$$

$$\alpha^2 + \beta^2 = 12 + 27 = 39$$

Q15

$$f(x) = -(p^2 - 6p + 8) \cos 4x + 2(2 - p)x + 7$$

$$f'(x) = +4(p^2 - 6p + 8) \sin 4x + (4 - 2p) \neq 0$$

$$\sin 4x \neq \frac{2p - 4}{4(p - 4)(p - 2)}$$

$$\sin 4x \neq \frac{2(p - 2)}{4(p - 4)(p - 2)}$$

$$p \neq 2$$

$$\sin 4x \neq \frac{1}{2(p - 4)}$$

$$\Rightarrow \left| \frac{1}{2(p - 4)} \right| > 1$$

on solving we get

$$\therefore p \in \left(\frac{7}{2}, \frac{9}{2} \right)$$

$$\text{Hence } a = \frac{7}{2}, b = \frac{9}{2}$$

$$\therefore 16ab = 252$$

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